

Class-AB Rail-to-Rail CMOS Analog Buffer

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Abstract— In this paper a low-power rail-to-rail CMOS analog buffer is presented. The circuit is based on an input stage made up of two complementary class AB differential pairs, while a simple additional circuit allows rail-to-rail operation at the output terminal. The proposed circuit combines low static power consumption and high drive capability, resulting very suitable for applications with large capacitive loads. The buffer has been designed in a 0.35- μm CMOS technology to operate with a ± 1.5 V dual supply. Simulated results are provided in order to demonstrate the proper operation of the proposed circuit. A rail-to-rail signal swing is achieved and a THD lower than -44 dB is obtained for a $2.4\text{-}V_{pp}$ 100-kHz input sinewave signal, whereas the input capacitance is lower than 32 fF.

I. INTRODUCTION

Analog voltage buffers are important building blocks in mixed signal designs, where they are used for testing or signal monitoring and for driving large capacitive loads [1-9]. In the first case, the buffer is usually connected to internal nodes of the circuit under test. Therefore, any increase of the parasitic capacitance at this node could be critical and the performance could be degraded. For this reason, an important feature in an analog buffer is to present a low input capacitance. On the other hand, when the buffer is used to drive loads the input capacitance is not a significant problem. However, a high slew-rate along with a large output signal swing is desirable in order to drive the load over the entire voltage range in a fast way.

The total supply voltage of present integrated circuits has been reduced, attending mainly to power consumption and reliability issues. This trend has forced most basic building blocks to be redesigned, in an attempt to operate under so stringent supply conditions. However, in an analog circuit a low supply voltage is not necessarily equivalent to a low power consumption, as the overall performance of analog blocks has to be guaranteed. In addition, noise levels remain unchanged in analog circuits while the available voltage range for the signal decreases. Therefore, rail-to-rail operation is essential in low-voltage designs, in order to maximize the signal-to-noise ratio.

Different techniques have been previously proposed to implement a buffer meeting the features pointed out above [1-9].

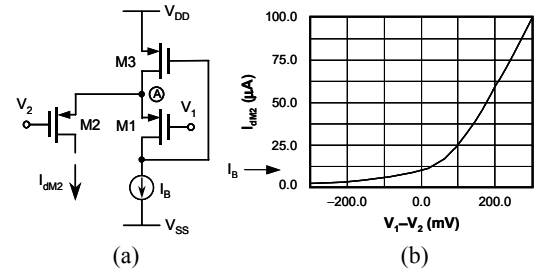


Figure 1. (a) Class AB differential input cell and (b) its DC transfer characteristic for a differential input signal.

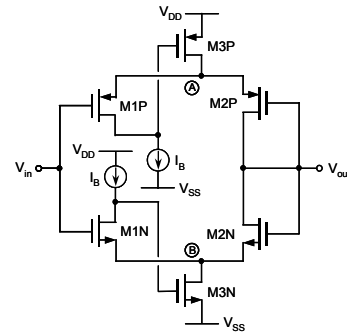


Figure 2. Low power buffer based on two complementary class AB differential input cells.

This paper introduces a circuit technique to obtain a rail-to-rail CMOS analog buffer with class AB behavior, which leads to a low power approach. Simulated results are provided to demonstrate the correct operation of the proposed solution.

II. PREVIOUS IMPLEMENTATION

A good approach based on using a complementary input stage with class AB behavior was introduced in [9]. Figure 1(a) illustrates a p-channel class AB differential pair, able to deliver a large amount of current when a large differential signal is applied to its input terminals [10]. The impedance

TABLE I
TRANSISTOR ASPECT RATIOS (IN $\mu\text{m}/\mu\text{m}$) FOR THE PROPOSED
RAIL-TO-RAIL CLASS-AB BUFFER.

M1P, M2P, M1PR	60/1
M1N, M2N, M1NR	20/1
M3P, M4N, M5N	30/1
M3N, M4P, M5P	10/1
M2PR-M5PR	10/1
M2NR-M5NR	30/1
MSP	1.8/1
MSN	0.6/1

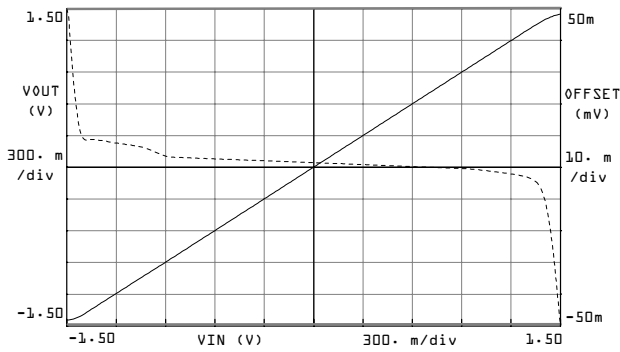


Figure 4. DC transfer characteristic of the analog buffer in Fig. 3 (solid line: output voltage, dashed line: offset voltage).

to the bottom part of the output branch, turning off the buffer. A similar situation arises close V_{SS} , where the n-channel differential pair is not active. For this reason, transistors M1PR-M5PR and M1NR-M5NR have been included in the circuit in Fig. 2, maintaining active the output branch over the entire voltage range.

Thus, the operation of the buffer can be detailed as follows. When the input signal, V_{in} , is in the midsupply region, the two input pairs, M1P-M2P and M1N-M2N, are active and M4P-M5P and M4N-M5N, respectively, mirror a current equal to I_B to the output branch. Furthermore, a replica of the current I_B is copied through transistors M1PR (M1NR) and M2PR-M3PR (M2NR-M3NR), providing the current required by the current source in the additional circuit at the bottom (top) of the output branch. As a consequence, transistors M4PR and M5PR (M4NR and M5NR) are turned off and do not contribute to the output current. When the input signal goes close to V_{DD} , the PMOS input pair cuts off and the replica of the inverting input branch, i.e., M1PR-M3PR, does not send any current contribution to the current source in the additional circuit at the output. Thus, M4PR and M5PR turn on, drawing a current equal to I_B from the output branch and, thus,

maintaining the buffer turned on. Something similar happens when the input signal, V_{in} , is close to V_{SS} and, therefore, rail-to-rail operation is reached at the input as well as at the output terminal of the circuit.

It is worth to note that voltage level-shifters have been included in the input stage so as to drive transistors M3P and M3N, avoiding the operation of M1P and M1N, respectively, in the linear region and extending the signal range at the input branch up to both supplies.

The dynamic operation of the proposed buffer is enhanced by the high drive capability of the class AB differential cells in the input branch of the circuit. In the case of a large input signal in the positive direction, transistor M2P cuts off whereas transistor M2N draws a large amount of current. This current is then mirrored to the output branch providing a copy of it to the load and allowing the output node to follow the input voltage. On the contrary, when a large input signal is applied in the negative direction, transistor M2N cuts off and M2P delivers a large current, which is drawn from the load when the output voltage follows the input signal.

The input capacitance of the buffer can be reduced by scaling down the size of transistors M2P and M2N with respect to the geometry of M1P and M1N. Nevertheless, it must be pointed out that the reduction of the aspect ratio of these transistors would lead to a decrease of their effective drive capability.

The bandwidth of the proposed buffer can be very large as there is only one high impedance node in the circuit. However, the single-gain-stage configuration adopted along with the high output impedance of the output node makes this configuration more suitable to be used for driving large capacitive loads, given that low resistive loads would reduce the total gain of the buffer and, consequently, its accuracy.

IV. SIMULATED RESULTS

The analog voltage buffer in Fig. 3 was implemented in a $0.35\text{-}\mu\text{m}$ CMOS technology, with nominal threshold voltages of 0.5 V and -0.6 V for NMOS and PMOS transistors, respectively. The circuit was designed to operate with a supply voltage of ± 1.5 V, a biasing current of $10\mu\text{A}$, and a capacitive load of 10 pF. The aspect ratios of the transistors in the circuit in Figure 3 are detailed in Table I.

Figure 4 shows the DC transfer characteristic of the proposed analog buffer along with the offset voltage. As expected, rail-to-rail behavior is achieved.

Figure 5 illustrates the large-signal transient response of the circuit in Fig. 3 to $2.4\text{-}V_{pp}$ $0.5\text{-}\mu\text{s}$ input pulses. Particularly, in Fig. 5(a) the input and output voltages are depicted. The circuit features a high slew-rate due to the class AB operation of the input stage. Figure 5(b) illustrates the currents flowing through the output transistors of the

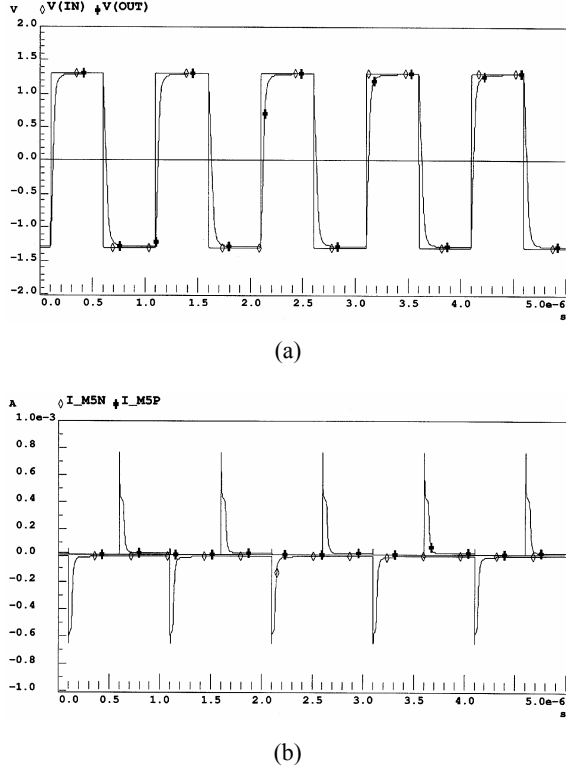


Figure 5. Large-signal transient response of the buffer in Fig. 3 with a 10-pF load to a 2.4-V_{pp}, 0.5-μs square input signal: (a) input and output voltages and (b) currents through the output transistors.

buffer. A large ratio between the maximum output current delivered by the buffer and the quiescent biasing current can be observed, demonstrating that the proposed solution leads to low power consumption.

Finally, Table II provides a summary of the performance of the proposed buffer. All the parameters in the table were obtained for an input voltage equal to zero. The simulated open-loop gain and unity-gain frequency were approximately 54 dB and 6.1 MHz. The relatively low value of the gain is due to the single-gain-stage arrangement of the circuit. The value of the GBW can be increased at the expense of increasing the biasing current of the input differential pairs and, so, raising the power consumption of the circuit. A total harmonic distortion (THD) equal to -44.6 dB is obtained for a 2.4-V_{pp} 100-kHz input sinewave signal. The simulated input capacitance of the proposed buffer is lower than 32 fF when no scaling is performed in the input transistors.

V. CONCLUSION

In this paper a rail-to-rail voltage buffer with reduced input capacitance has been proposed. Its class-AB operation provides a low-power consumption, resulting very suitable to drive large capacitive loads. Input rail-to-rail operation is

TABLE II
SIMULATED PERFORMANCE OF THE ANALOG BUFFER IN FIGURE 3
($V_{DD} = -V_{SS} = 1.5$ V, $I_B = 10$ μA, $C_L = 10$ pF).

Parameter	Value
Open-loop gain	54 dB
Unity-gain frequency	6.1 MHz
Phase margin	86 °
Power dissipation	0.33 mW
Input capacitance	31.5 fF
THD (1.0 V _{p-p} @ 100 kHz)	-54.2 dB
(2.4 V _{p-p} @ 100 kHz)	-44.6 dB
SR ⁺ / SR ⁻	56 / 44 V/μs

achieved by connecting two complementary differential pairs in parallel, while the output voltage can also be swept over the entire input voltage range, thanks to an additional circuit connected to the output branch. Simulated results have been provided in order to demonstrate that the proposed circuit can be used to test analog signals and to drive large capacitive loads.

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